MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL COAXIAL, RADIO FREQUENCY (SERIES BNC, SOLDER POCKET, SOCKET CONTACT JAM NUT MOUNTED, ISOLATED, CLASS 2)

This specification is approved for use within all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.

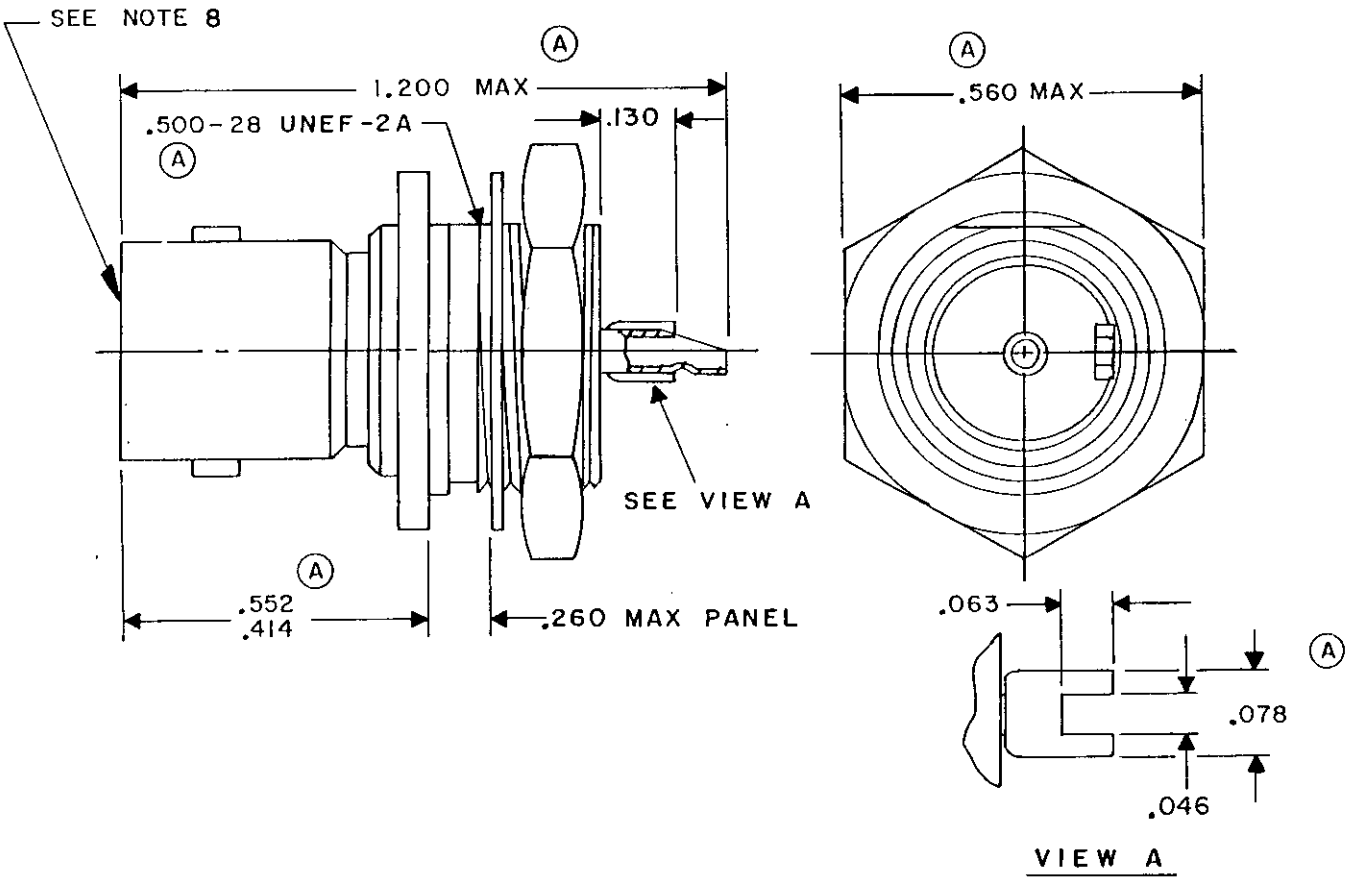
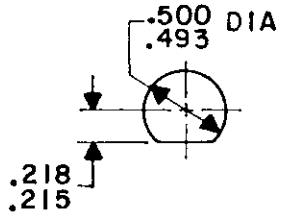


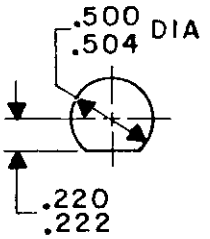
FIGURE 1. General configuration.

(A) denotes changes

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CONNECTOR "D" MTG SURFACE



MOUNTING HOLE

M39012/128-0001

Inches	mm	Inches	mm
.046	1.17	.222	5.64
.063	1.60	.260	6.60
.078	1.98	.414	10.52
.130	3.30	.493	12.52
.215	5.46	.500	12.7
.218	5.54	.504	12.80
.220	5.59	.552	14.02
		.560	14.22

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
4. Dimensions A defines the overall length of connector when assembled to the cable.
5. All undimensioned pictorial configurations are for reference purposes only.
6. Unless otherwise specified, all tolerances are ±.005 inch (0.13 mm).
7. Solder pocket shall accommodate a wire of .059 (1.50 mm) diameter maximum.
8. Series BNC, socket contact interface in accordance with MIL-STD-348.

FIGURE 1: General configuration - Continued.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 4,000 MHz.

Voltage rating: 500 volts rms, maximum working voltage at sea level. 125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and construction: See figure 1.

Center contacts:

Axial force: 4 lbs, minimum.
Radial torque: 4 inch-ounces, minimum.

Isolating insulator captivation: Forces shall be applied between the mounting bushing and connector body for a period of 10 seconds, minimum.

Torque: 15 inch-pounds.
Axial force: 30 pounds.

Force to engage and disengage:

Longitudinal force: 3 pounds, maximum.
Torque: 2-1/2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions (coupling torque): Not applicable.

Ⓐ Mating characteristics: Reference MIL-STD-348.

Center contact (socket):

Oversize test pin: .057 diameter, minimum (nonclosed entry contacts only).
Insertion depth: .125, minimum.
Number of insertions: One.

Insertion force test: Steel test pin diameter .054, minimum.

Test pin finish: 16 microinches.
Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .052, maximum.

Withdrawal force: 2 ounces, minimum.
Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B of MIL-STD-202. 5,000 megohms minimum.

Contact resistance (In milliohms maximum):

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable

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Resistance to test prod change: Not applicable.

Corrosion (salt spray): Method 101, test condition B of MIL-STD-202.

Voltage standing wave ratio (VSWR): Not applicable.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 1,500 volts rms, minimum at sea level.

Corona level:

Voltage: 375 volts, minimum.
Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms, at a frequency between 5 and 7.5 MHz.
Leakage current: Not applicable.

Connector durability: 500 cycles minimum, at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Vibration, high frequency: Method 204, test condition B of MIL-STD-202. No discontinuity permitted.

Shock: Method 213, test condition G of MIL-STD-202. No discontinuity permitted.

Thermal shock: Method 107, test condition B of MIL-STD-202, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity. Dielectric withstanding voltage shall be met. Voltages shall be applied at points suitable for testing each individual insulator.

Cable retention force: Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

Isolation: The following tests shall be accomplished after the shock test to verify isolation. Voltages shall be applied at two points suitable for testing only the insulator providing the isolation.

a. Insulation resistance, method 302, test condition B of MIL-STD-202. 5,000 megohms, minimum.

b. Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms, minimum.

Part number: M39012/128-0001.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3566)

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